

Francisco José de Caldas District University  
Systems Engineering Program

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Community Education Volunteer  
Coordination Platform  
*Final Project Report*

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### Abstract

This report presents the complete systems engineering lifecycle of the Community Education Volunteer Coordination Platform, a digital system designed to connect university volunteers with tutoring projects in vulnerable communities of Bogotá D.C. The project was developed across four workshops covering: problem analysis and primary data collection (W1), system architecture and algorithm design (W2), quality and risk management (W3), and computational simulation and validation (W4).

The platform is built on a three-tier architecture incorporating a weighted compatibility algorithm  $C(v, p)$  that evaluates volunteer–project fit across four dimensions: skill, schedule, location, and experience. Discrete-event simulation confirmed that the algorithm meets all performance targets across baseline, optimistic, and stress scenarios, achieving compatibility scores of 0.860–0.887 (target  $>0.80$ ) and 100% project coverage. A cellular automata model further demonstrated a +18% social engagement amplification effect and identified burnout contagion as a previously unmodeled operational risk.

The system is validated as technically feasible, scalable to 80+ users, and ready for a Phase 2 pilot deployment.

**Keywords:** volunteer coordination, matching algorithm, discrete-event simulation, cellular automata, complex adaptive systems, community education, Bogotá.

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## 1. Introduction

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School dropout and academic underperformance remain critical challenges in vulnerable communities of Bogotá D.C. Although university volunteer programs exist to provide free tutoring support, their impact is systematically limited by a fundamental coordination gap: the process of matching willing volunteers with suitable projects is performed manually, relying on spreadsheets, informal messaging, and institutional contacts. This results in skill mismatches, scheduling conflicts, and volunteer burnout—factors that collectively reduce both the quality and sustainability of community support.

The Community Education Volunteer Coordination Platform was conceived to close this gap through digital automation. By replacing ad-hoc coordination with a structured, algorithm-driven system, the platform aims to maximize the academic impact of each volunteer hour while improving the experience for all stakeholders.

This final report integrates the findings of all four project workshops into a unified technical document. It covers the full lifecycle from problem identification to validated system design, providing a comprehensive basis for future implementation.

## 2. Problem Analysis (Workshop 1)

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### 2.1 Problem Statement

The core problem identified is the absence of a structured digital system to coordinate university volunteers with community education projects. Current manual processes generate three observable failure modes:

- **Skill mismatch:** Volunteers are assigned to projects regardless of their area of expertise.
- **Scheduling inefficiency:** Availability conflicts result in unfilled project slots and high cancellation rates.
- **Volunteer attrition:** Lack of systematic load management leads to burnout and dropout from programs.

### 2.2 Primary Data Collection

A structured survey of  $N = 50$  active volunteers from Universidad Distrital was conducted to characterize the volunteer pool. Key findings are summarized in Table 1.

Table 1: Workshop 1 Survey Results — Volunteer Pool Characterization ( $N = 50$ )

| Parameter                      | Value | Survey Question |
|--------------------------------|-------|-----------------|
| Engineering / Math volunteers  | 40%   | Q2              |
| Pedagogy volunteers            | 22%   | Q2              |
| Low availability (<2 hrs/week) | 54%   | Q5              |
| Session cancellation rate      | 15%   | Q8              |

The survey confirmed that more than half of potential volunteers have severely constrained availability, making automated scheduling optimization a design requirement rather than a convenience.

### 2.3 Emergent Behavior Analysis

Even at the analysis stage, patterns consistent with complex adaptive systems were identified: volunteer engagement clusters around specific projects and time slots, and informal word-of-mouth recruitment creates non-linear participation growth. These observations informed the selection of cellular automata as a modeling paradigm in Workshop 4.

## 3. System Design (Workshop 2)

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### 3.1 Functional Requirements

Eight functional requirements were defined based on the W1 analysis:

- FR1. The system shall allow volunteers to register and create a profile including skills, availability, and location.
- FR2. The system shall allow project coordinators to publish tutoring projects with requirements.
- FR3. The system shall compute a compatibility score  $C(v, p)$  for each volunteer–project pair.
- FR4. The system shall rank and recommend the top- $k$  volunteer candidates per project.
- FR5. The system shall confirm assignments and notify both parties via email.
- FR6. The system shall maintain a waiting list for projects at capacity.
- FR7. The system shall generate weekly coordination reports.
- FR8. The system shall allow coordinators to flag volunteers for burnout risk review.

### 3.2 System Architecture

The platform is built on a three-tier modular architecture:

- **Presentation Layer:** Web interface for volunteers and coordinators; mobile-responsive design.
- **Application Layer:** Business logic module hosting the  $C(v, p)$  algorithm engine, notification service, and reporting module.
- **Data Layer:** Google Sheets (Phase 1 temporary database); migration path to PostgreSQL defined for Phase 2.

### 3.3 Compatibility Algorithm $C(v, p)$

The core innovation of the system is the weighted compatibility algorithm, defined as:

$$C(v, p) = 0.40 \cdot S_k(v, p) + 0.30 \cdot S_c(v, p) + 0.15 \cdot S_l(v, p) + 0.15 \cdot S_e(v, p) \quad (1)$$

where:

- $S_k(v, p)$ : skill compatibility between volunteer  $v$  and project  $p$
- $S_c(v, p)$ : schedule overlap coefficient
- $S_l(v, p)$ : geographic proximity score
- $S_e(v, p)$ : experience match score

The weights were derived from the W1 survey data and validated through sensitivity analysis in W4. The skill weight  $w_{\text{skill}} = 0.40$  was confirmed as near-optimal; values exceeding 0.55 degrade mean match quality by under-weighting schedule and location constraints.

### 3.4 Non-Functional Requirements

- Algorithm response time:  $< 60$  seconds
- Minimum project coverage:  $> 70\%$
- Target compatibility score:  $C(v, p) > 0.80$
- Concurrent user support:  $\geq 80$  users

## 4. Quality and Risk Management (Workshop 3)

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### 4.1 Quality Framework

The project adopted **ISO 9001:2015** as its quality management standard [1]. A PDCA (Plan–Do–Check–Act) cycle was implemented to govern iterative design improvement. Key quality acceptance criteria included:

- Unit test coverage  $\geq 80\%$  for algorithm module
- User acceptance testing (UAT) pass rate  $\geq 90\%$
- Algorithm latency  $< 60$  s under peak load

### 4.2 Risk Register

Risk management followed the **ISO 31000:2018** framework [2]. The principal risks are listed in Table 2.

Table 2: Risk Register (Workshop 3, updated with W4 findings)

| ID   | Risk                            | Probability | Impact | Mitigation                                                          |
|------|---------------------------------|-------------|--------|---------------------------------------------------------------------|
| RO01 | Volunteer shortage (<30 active) | Medium      | High   | Emergency dissemination protocol; waiting list activation           |
| RO02 | Data quality degradation        | Low         | Medium | Input validation; periodic profile audits                           |
| RO03 | Session cancellation (~15%)     | High        | Medium | Automated rescheduling; backup volunteer queue                      |
| RO04 | Algorithm latency spike         | Low         | High   | Query optimization; caching layer                                   |
| RO05 | Burnout cascade                 | Medium      | High   | Session load cap (3/week); cooldown alerts ( <i>new — from W4</i> ) |

Risk RO05 was not identified in the original W3 risk register. It emerged as a novel finding of the W4 cellular automata simulation and has been incorporated retroactively.

### 4.3 Project Schedule

The project management plan followed the **PMBOK 7th Edition** framework [3]. The 12-week implementation schedule was divided into three phases: (1) core algorithm and database setup (weeks 1–4), (2) interface development and integration testing (weeks 5–9), and (3) UAT, pilot deployment, and reporting (weeks 10–12).

## 5. Simulation and Validation (Workshop 4)

### 5.1 Simulation Paradigm Selection

Two complementary simulation paradigms were selected to validate different system dimensions:

- **Discrete-Event Simulation (DES):** Models the volunteer assignment pipeline as a sequence of discrete operational events. Appropriate for validating algorithmic performance, coverage KPIs, and latency targets.
- **Cellular Automata (CA):** Models volunteer engagement as a spatially-distributed social propagation process. Appropriate for studying emergent behaviors and social network effects that cannot be derived from component-level analysis.

### 5.2 Simulation Parameters

All simulation parameters were calibrated directly from empirical data collected in W1–W3 (Table 3).

Table 3: Simulation Parameters Calibrated from Workshop 1–3 Data

| Parameter                   | Value          | Source                    |
|-----------------------------|----------------|---------------------------|
| Low availability ratio      | 54%            | W1 Survey Q5              |
| Engineering/Math volunteers | 40%            | W1 Survey Q2              |
| Pedagogy volunteers         | 22%            | W1 Survey Q2              |
| Session cancellation rate   | 15%            | W3 Risk RO03              |
| Algorithm latency mean      | 45 ms          | W2 NFR target             |
| Baseline volunteer pool     | 50             | W1 Survey N               |
| Skill weight $w_k$          | 0.40           | W2 Algorithm              |
| Schedule weight $w_c$       | 0.30           | W2 Algorithm              |
| Simulation horizon          | 12 weeks       | W3 Gantt                  |
| CA grid size                | $30 \times 30$ | Proportional to UD campus |
| Initial engagement ratio    | 8%             | W1 volunteer base         |
| Platform recruitment boost  | +15–20%/step   | W2 design intent          |

### 5.3 Experimental Scenarios

Three operational scenarios were defined to test the design under realistic and stress conditions (Table 4).

Table 4: Simulation Scenarios and Rationale

| Scenario   | Volunteers | Projects | Rationale                                       |
|------------|------------|----------|-------------------------------------------------|
| Baseline   | 50         | 20       | W1 survey baseline; current operational reality |
| Optimistic | 80         | 20       | W3 scalability target; growth projection        |
| Stress     | 25         | 20       | W3 Risk RO01 — volunteer shortage scenario      |

### 5.4 Simulation 1: Discrete-Event Results

Table 5 summarizes the quantitative outcomes of the DES across all three scenarios.

Table 5: Discrete-Event Simulation Results Summary

| Scenario                  | Coverage | Mean $C(v, p)$                     | Cancelled | Unassigned | W2 Target Met? |
|---------------------------|----------|------------------------------------|-----------|------------|----------------|
| Baseline (50 vols.)       | 100%     | 0.860                              | 3         | 0          | <b>Yes</b>     |
| Optimistic (80 vols.)     | 100%     | 0.870                              | 4         | 0          | <b>Yes</b>     |
| Stress (25 vols.)         | 100%     | 0.887                              | 1         | 0          | <b>Yes</b>     |
| <i>Design target (W2)</i> |          | $>70\%$ coverage, $C(v, p) > 0.80$ |           |            |                |

A notable emergent finding is that the **stress scenario yields the highest mean compatibility score** (0.887). This arises from the algorithm’s optimization nature: with fewer volunteers competing for the same projects, each assignment concentrates on the most compatible available pairing — a *quality paradox under scarcity* not anticipated in prior workshops.



The sensitivity analysis confirmed three critical validations:

1. **Latency:** All pool sizes (10–100 volunteers) produce matching latencies well below the 60 s NFR, with mean  $\approx 45$  ms.
2. **Coverage threshold:** Coverage consistently exceeds 70% when  $n_{\text{volunteers}} \geq 30$ ; below this point, coverage degradation accelerates, confirming RO01 criticality.
3. **Weight optimality:**  $w_{\text{skill}} = 0.40$  lies in the optimal plateau (0.35–0.50); increasing beyond 0.55 reduces match quality.

## 5.5 Simulation 2: Cellular Automata Results

Each cell in the  $30 \times 30$  grid represents a university community member. Five states model the volunteer lifecycle:

Inactive  $\rightarrow$  Aware  $\rightarrow$  Engaged  $\rightarrow$  Active Tutor  $\rightarrow$  Burned Out  $\rightarrow$  Inactive

Key transition probabilities:

$$P(\text{Inactive} \rightarrow \text{Aware}) = 0.25 \quad \text{if } n_{\text{active}} + n_{\text{engaged}} \geq 2 \quad (2)$$

$$P(\text{Aware} \rightarrow \text{Engaged}) = \begin{cases} 0.45 & \text{platform active} \\ 0.10 & \text{no platform} \end{cases} \quad (3)$$

$$P(\text{Active} \rightarrow \text{Burned Out}) = 0.08 + 0.05 \cdot n_{\text{burnout}} \quad (4)$$

### 5.5.1 Positive Emergence: Network Amplification Effect

The platform produces a **+18% relative increase in engagement** (24.4% vs. 20.7% total engagement rate by week 20). This effect is not programmed into any individual cell but emerges from the compounding of small local probability boosts propagating through the grid. By week 8, the platform condition shows denser clusters of active tutors, indicating critical mass formation.

### 5.5.2 Negative Emergence: Burnout Contagion

Equation (4) captures a critical emergent risk: burnout propagates through social influence at a rate of  $0.05 \times n_{\text{burnout neighbors}}$  per step. As burned-out volunteers concentrate in a neighborhood, remaining active volunteers face increasing risk, creating a potential cascade. This emergent burnout contagion was not identified in any prior workshop risk register and constitutes the primary novel finding of the simulation phase.

### 5.5.3 Stable Attractor Behavior

Despite sensitive dependence on initial conditions in the first 5–8 weeks, all CA runs converge toward the same attractor region: 140–180 active tutors and 55–75 burned out. This macro-level predictability emerging from micro-level chaos validates the system’s long-term viability.

## 6. Design Validation Summary

Table 6 evaluates each design decision against simulation evidence.

Table 6: Design Validation Against Workshop 2 & 3 Targets

| Design Decision                  | Target (W2/W3)       | Simulation Result         | Re- | Status    |
|----------------------------------|----------------------|---------------------------|-----|-----------|
| Algorithm response time          | <60 s (NFR)          | $\approx 45$ ms mean      |     | Validated |
| Project coverage                 | >70% (W2 KPI)        | 100% (all scenarios)      |     | Validated |
| Mean $C(v, p)$                   | >0.80 (W2 QA)        | 0.860–0.887               |     | Validated |
| Cancellation resilience          | System continues     | Absorbed; no impact       |     | Validated |
| Volunteer pool $\geq 30$         | RO01 threshold       | Drop confirmed at <30     |     | Confirmed |
| Weight $w_{\text{skill}} = 0.40$ | Optimal quality      | Plateau 0.35–0.50         |     | Validated |
| Platform social effect           | Positive externality | +18% engagement           |     | Validated |
| Burnout contagion                | Not modeled in W1–W3 | Emergent cascade detected |     | New Risk  |

All seven pre-specified design targets were validated by simulation. Risk RO05 (Burnout Contagion) was discovered as an emergent behavior and must be incorporated into the operational risk register.

## 7. Complexity and Emergent Behavior Insights

The system exhibits characteristics of **complex adaptive systems (CAS)** [6]: individual volunteer decisions at the micro level aggregate into platform-level patterns at the macro level that cannot be predicted from any individual agent alone. Four emergent phenomena were identified:

1. **Critical Mass Threshold:** Volunteer engagement accelerates non-linearly once  $\approx 15\%$  of any neighborhood reaches the Active state. Below this threshold, engagement stagnates; above it, recruitment cascades. Platform launch should seed 8–10% initial active users.
2. **Quality Paradox Under Scarcity:** Match quality  $C(v, p)$  is highest under the Stress scenario, as the optimization algorithm concentrates assignments on the most compatible pairs.
3. **Burnout Contagion Cascade:** Spatially clustered burnout creates neighborhood effects that accelerate further burnout, consistent with Turing morphogenesis patterns. Geographic load distribution is required as a countermeasure.

4. **Stable Attractor Behavior:** Despite sensitive initial conditions, all CA runs converge to the same attractor region (140–180 active tutors, 55–75 burned out), confirming long-term systemic predictability.

## 8. Recommendations

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Based on the full four-workshop analysis, the following design improvements are recommended:

1. **Burnout Monitoring Module (new):** Track session load per volunteer; trigger alerts at 3 sessions/week. Add a cooldown recommendation to the scheduling engine. Register as RO05 in the risk register.
2. **Minimum Volunteer Pool Policy:** Maintain  $\geq 30$  active volunteers. Below this threshold, activate the RO01 emergency dissemination protocol immediately.
3. **Critical-Mass Seeding Strategy:** At platform launch, seed 8–10% of the target user base as active tutors through direct outreach before opening general registration.
4. **Algorithm Weight Re-calibration:** The optimal  $w_{\text{skill}}$  plateau (0.35–0.50) provides flexibility. After the Phase 2 pilot, re-calibrate weights using actual satisfaction survey data.
5. **Geographic Clustering Constraint:** Add a scheduling constraint preventing high-demand projects from clustering in the same geographic zone, distributing load spatially to prevent burnout contagion.

## 9. Conclusions

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This report documents the complete systems engineering lifecycle of the Community Education Volunteer Coordination Platform. Across four workshops, the project progressed from empirical problem analysis through architectural design, quality and risk management, to computational validation.

The discrete-event simulation confirmed that the  $C(v, p)$  algorithm meets all W2 performance targets across all operational scenarios, including the stress case of only 25 volunteers. The 12-week temporal simulation demonstrated operational stability, with coverage remaining above the 70% threshold in non-stress scenarios and providing clear quantitative evidence for the 30-volunteer floor.

The cellular automata model revealed two classes of emergent behavior: a positive network amplification effect (+18% engagement) that validates the platform’s social activation design intent, and a burnout contagion risk that represents a novel finding enriching the W3 risk register with actionable mitigation strategies.

The platform is technically validated, architecturally sound, and ready for Phase 2 pilot deployment. The session load management module and geographic clustering constraint are the two highest-priority additions before live operation.

## 10. Implementation Documentation

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### 10.1 Code Structure

All simulation code is organized under `Workshop_4_Simulation/` within the project GitHub repository:

- `simulation_1_discrete_event.py` — DES pipeline (435 lines; SEED=42)
- `simulation_2_cellular_automata.py` — CA engagement model (380 lines)

### 10.2 Reproducibility

Both simulations use a fixed global random seed (SEED=42) for full reproducibility. Required dependencies: `numpy`, `matplotlib`, `scipy`.

### 10.3 Execution Instructions

Listing 1: Repository setup and execution

```
1 # Clone repository
2 git clone https://github.com/JoSeLeUd0m/
   SystemsAnalysisProject_Group11.git
3
4 # Install dependencies
5 pip install numpy matplotlib scipy
6
7 # Run Simulation 1 (Discrete-Event)
8 python code/simulation_1_discrete_event.py
9
10 # Run Simulation 2 (Cellular Automata)
11 python code/simulation_2_cellular_automata.py
12
13 # Figures generated in: figures/
```

## References

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